Name of Student : Raghav Mahesh Sarda	**PRN:** 25070122***235***
Subject: Programming with Java	**Semester: III**
ASSIGNMENT No : 1	**Year: AY 2026-27**
TITLE : Java Environment Setup and Basic Program Execution	

Problem Statement

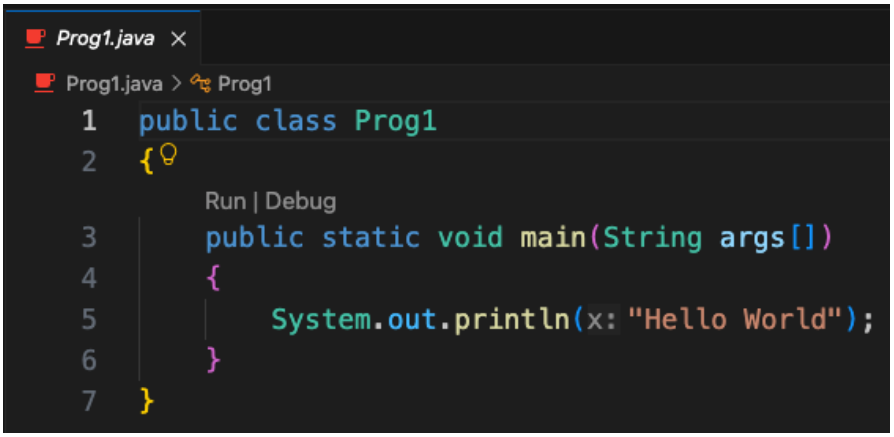
Develop an application based on Java environment setup, JDK/JRE/JVM, variables, methods and constructors. and demonstrate the implementation of the concepts through suitable examples.

Learning Objectives

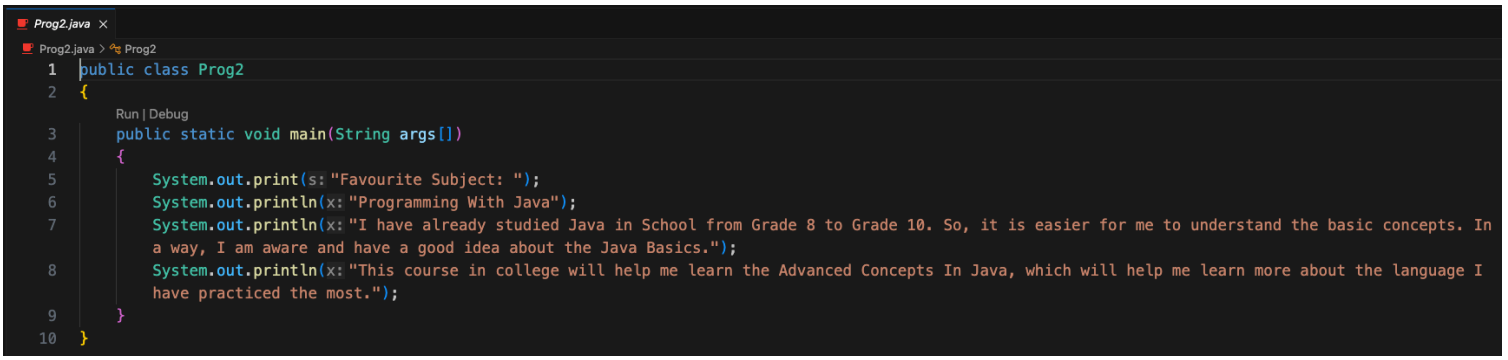
Understand the theoretical concepts, implement them in Java programs, analyse outputs, and apply the concepts to solve practical programming problems.

Programs

Prog1.java

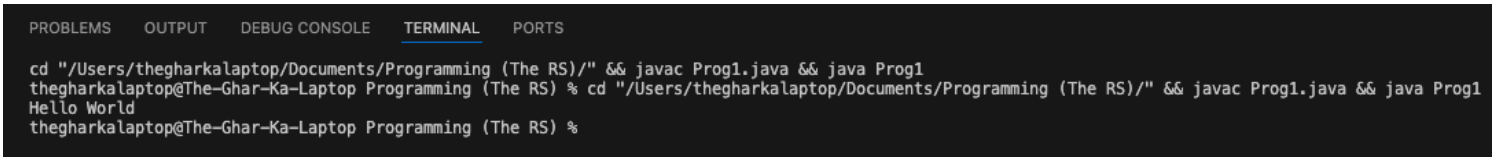


Prog2.java

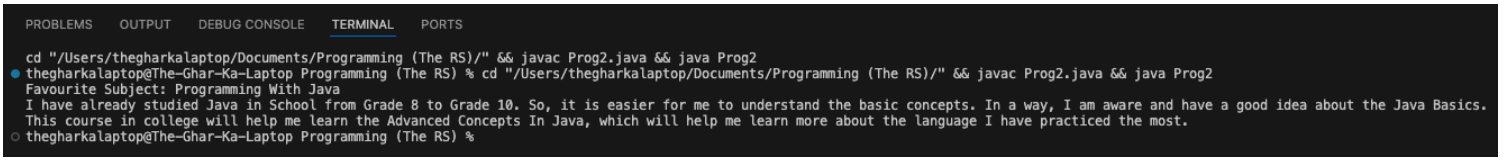


Outputs

Prog1.java



Prog2.java



Conclusion

In this assignment, the foundational concepts of Java programming, including JDK/JRE/JVM architecture, environment setup, variables, methods, and constructors, were successfully studied and implemented. Demonstrating variables, methods, and constructor initialization provided clarity on dynamic data storage, code reusability, and object-oriented structure.

Questions

1. Explain the main concept used in Java environment setup, variables and methods.
2. What are the advantages of using Java variables and methods?
3. How can this implementation be improved?
4. What are the real-world applications?

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Q1) Explain the main concept used in Java environment setup, variables and methods.

Ans: Java environment setup relies on installing the Java Development Kit (JDK), setting path variable, and using the JVM to execute compiled bytecode across platforms.
Variables act as named containers in memory to store specific types of data during program execution. Methods are reusable blocks of code designed to perform specific tasks and handle application logic.

Q2) What are the advantages of using Java variables and methods?

Ans: Variables allow dynamic data manipulation and efficient memory utilization by holding changing values.
Methods promote code reusability, reduce redundancy, and make programs modular and easier to debug.
Together, they improve application structure, readability and maintainability.

Q3) How can this implementation be improved?

Ans: The code structure can be improved by adding input validation to prevent runtime errors during user input. Incorporating Object-Oriented Programming (OOPs) principles such as encapsulation and constructors enhances data safety and object initialization. Additionally, implementing proper exception handling ensures smooth execution without abrupt program crashes.

Q4) What are the real-world applications?

Ans: These foundational concepts are used in building desktop software, web backends and mobile applications. They power basic modules in enterprise systems like banking software, employee management systems and e-commerce platforms. Every production-level Java application uses variables and methods for core data handling and business logic processing.